THE LAWS OF SOFTWARE TEAMS

Kernighan's law

Debugging is twice as hard as writing the code

swipe →

Ivan Rykov

01 THE HOOK

This line is clever
Nobody can fix it

billing.py · line 212

```
v = (x>>3^y<<1)&~m|k and f(*z[::-1])
```

ALERT · 03:12 · six months later
Invoices are failing



on call tonight
nobody can read it

illustration

02 THE LAW

Debugging is twice as hard

Brian Kernighan & P. J. Plauger · 1978

> “Everyone knows that debugging is twice as hard as writing a program in the first place. So if you’re as clever as you can be when you write it, how will you ever debug it?”

The Elements of Programming Style, 2nd edition, 1978, chapter 2

Code written at your limit
is code you cannot debug

Spend only half your cleverness



illustration of Kernighan's argument

04 THE DATA

Code is read more than written

FIELD STUDY · 78 developers · 7 projects
3,148 working hours observed

58%

of their time went into understanding code



understanding — everything else · 42%

Xia et al., IEEE Transactions on Software Engineering, 2018

The writer pays once
Readers pay again

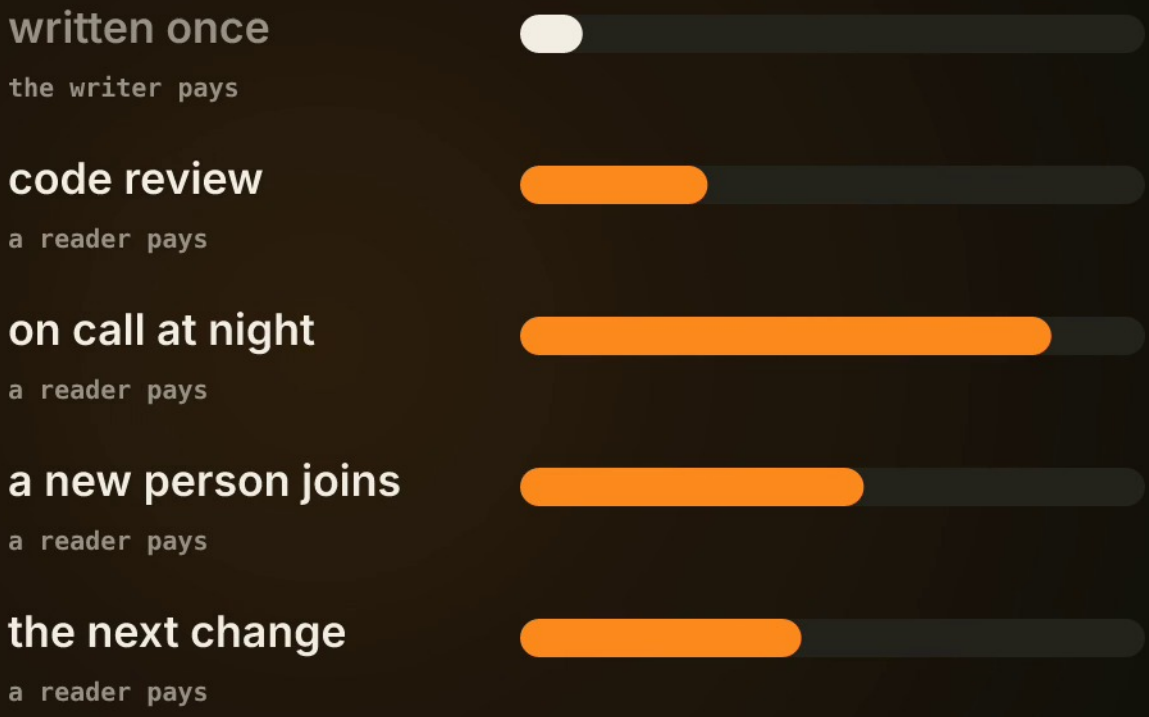


illustration · bars show relative effort

You get what you reward

CODE REVIEW

"Clever trick, nice!"

→ next month: more clever tricks

PROMOTION CASE

"Designed a complex system"

→ next year: more complex systems

Praise code that is easy to change

illustration

Make simple the standard

1. **Review for the person on call**
   they should be able to fix it at night
2. **Clever code: comment and test**
   explain why, and prove that it works
3. **Prefer boring patterns**
   plain names, standard tools
4. **Praise deletions**
   less code is less to debug

08 TAKEAWAY

Write for the person who debugs it

billing.py · line 212, rewritten

```
if invoice.is_overdue():
    send_reminder(invoice)
# boring, and easy to fix
```

Often, that person is you

TAKEAWAY

Write for the person who debugs it

Sources: Kernighan and Plauger, The Elements of Programming Style, 1978 · Xia et al., IEEE TSE, 2018

next: Chesterton's fence